CSA47 – Deep Learning

CO1 – ASSESSMENT 2: ANALYTICAL PROBLEM SOLVING

CO1: *Mathematical foundations of deep learning – linear algebra, probability and information theory, numerical computation, machine learning basics, capacity, overfitting and underfitting, estimation, bias and variance, Bayesian statistics, stochastic gradient descent, challenges motivating deep learning.*

Assessment Title: Analyse the Mathematics of a Diabetic Retinopathy Screening Model (Total Marks - 50)

Nature of Assessment: Individual (In-class) Duration: 2 Hours Bloom's Level: K4 – Analyse Date: 12.10.26

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Scenario — "RetinaCare" Diabetic Retinopathy Screening

RetinaCare is a tele-ophthalmology service that screens fundus images of diabetic patients in rural primary health centres. Before the model is approved for clinical use, the hospital's AI audit committee has asked you to analyse the mathematics behind its feature compression, loss function, optimiser and error behaviour using the data below.

Support every conclusion with the formula used and the intermediate steps, and give answers to four decimal places. Use log base 2 for information-theoretic quantities. A scientific calculator is permitted; programming tools are not permitted.

D1 Feature matrix: $A = \begin{vmatrix} 3 & 0 \\ 4 & 5 \end{vmatrix}$

D2 Loss $J(w) = \frac{1}{2} w^T H w - b^T w$, $H = \begin{vmatrix} 4 & 1 \\ 1 & 4 \end{vmatrix}$, $b = \begin{vmatrix} 5 \\ 5 \end{vmatrix}$, $w_0 = \begin{vmatrix} 0 \\ 0 \end{vmatrix}$

D3 Classes (No DR, Mild, Moderate, Severe)
True distribution $p = [0.5, 0.25, 0.125, 0.125]$
Untrained model $q_1 = [0.25, 0.25, 0.25, 0.25]$
Partly trained $q_2 = [0.4, 0.3, 0.2, 0.1]$

D4 Prevalence = 10%, Sensitivity = 90%, Specificity = 95%

D5 True score $y = 10$, noise variance $\sigma^2 = 1$
Model P predictions = [8, 9, 8, 9] Model Q predictions = [12, 7, 11, 8]

D6 Confusion matrix (100 images)

	Pred. Referable	Pred. Non-referable
Actual Referable	40	10
Actual Non-referable	5	45

Design Challenge:

RetinaCare will be deployed in a district where the prevalence of referable retinopathy is only 2%, and a missed referable case is considered ten times as costly as an unnecessary referral.

Analyse how this changes your conclusions and determine whether the screening policy should be retained, modified or redesigned.

TASK 1 – Break Down the Audit Problem [K4] **3 Marks**

Decompose the audit into its mathematical components using data items D1–D6.

Discuss:

- The mathematical concept behind each data item
- How each item relates to a stage of the model (representation, loss, optimisation, decision, evaluation)
- Which items depend on each other, and the assumptions you make

Evidence Expected

A structured decomposition linking each data item to its concept and model stage.

TASK 2 – Examine the Compressibility of the Feature Matrix [K4] **6 Marks**

Using D1, examine how much of the information in A is captured by its dominant direction.

Discuss:

- The singular value decomposition $A = U\Sigma V^T$, derived through A^TA
- How the two singular values compare, and what their ratio reveals
- How the rank-1 approximation A_1 differs from A, measured by $\|A - A_1\|_F$
- The percentage of energy retained, and whether rank-1 compression is justified

You must show the **complete derivation**; stating only the singular values will not receive full credit.

Evidence Expected

A complete SVD with an analysis of the energy distribution and the rank-1 trade-off.

TASK 3 – Analyse the Curvature of the Loss Surface [K4] **5 Marks**

Using D2, analyse what the Hessian reveals about the loss surface.

Discuss:

- The eigenvalues and eigenvectors of H, and the directions of high and low curvature
- Whether H is positive definite, and what this implies about the convexity of J(w) and uniqueness of the minimum
- How the condition number of H will influence gradient descent
- The minimiser $w^* = H^{-1}b$, verified by $\nabla J(w^*) = 0$

Evidence Expected

A curvature analysis linking eigenvalues, convexity and conditioning to optimiser behaviour.

TASK 4 – Differentiate the Models using Information Theory [K4] **5 Marks**

Using D3, analyse how far each model's predictions are from the true distribution.

Discuss:

- $H(p)$, $H(q_1)$ and the cross-entropy $H(p, q_1)$
- How $H(p, q_1)$ splits into $H(p)$ and $KL(p \parallel q_1)$, verified numerically
- How $KL(p \parallel q_1)$ and $KL(p \parallel q_2)$ differ, and which classes contribute most to each divergence
- What a divergence of zero would indicate

Evidence Expected

A verified decomposition of cross-entropy and a class-wise analysis of the divergence of each model.

TASK 5 – Analyse the Screening Decision using Bayes' Rule [K4] **4 Marks**

Using D4, examine what a positive or negative result actually tells the ophthalmologist.

Discuss:

- The probability tree for the screened population
- P(Referable | Positive) and P(Referable | Negative)
- Why P(Referable | Positive) is much lower than the sensitivity
- Whether a positive result alone is sufficient grounds for treatment

Evidence Expected

Correct posteriors with an analysis of why they differ from sensitivity and specificity.

TASK 6 – Trace the Convergence of Gradient Descent [K4] **5 Marks**

Using D2, trace how gradient descent approaches the minimiser.

Discuss:

- The gradient $\nabla J(w)$ and three iterations with $\eta = 0.1$, reporting w and J(w) after each step
- How the distance to w^* changes from one iteration to the next
- The pattern in the error reduction, and how it relates to the eigenvalues of H

Evidence Expected

Accurate iterations with an analysis of the convergence pattern linked to the eigenvalues of H.

TASK 7 – Differentiate Momentum from Plain Gradient Descent [K4] **4 Marks**

Extend Task 6 to compare optimiser behaviour.

Discuss:

- Two iterations with momentum ($\beta = 0.9$, $\eta = 0.1$) compared with plain gradient descent
- Whether momentum overshoots w^*, and what causes it
- The learning-rate bound for convergence, derived from the largest eigenvalue of H
- Why $\eta = 0.5$ fails, traced through the update

Evidence Expected

A comparison of the two optimisers and a derived stability bound with the failure case explained.

TASK 8 – Decompose the Prediction Error [K4] **4 Marks**

Using D5, break down the expected error of Models P and Q.

Discuss:

- The bias², variance and expected squared error (Bias² + Variance + σ^2) of each model
- Which error component dominates each model, and why
- Which model is preferable, based on the decomposition

Evidence Expected

A correct decomposition with the dominant error source of each model identified.

TASK 9 – Examine the Error Profile of the Classifier [K4] **3 Marks**

Using D6, examine where the screening model makes mistakes.

Discuss:

- Accuracy, error rate, precision, recall (sensitivity), specificity and F1-score
- Which type of error is more frequent, and its clinical consequence
- Why accuracy alone gives an incomplete picture

Evidence Expected

Correct metrics with an analysis of the error types and their consequences.

TASK 10 – Relate the Findings to the Audit Question [K4] **3 Marks**

Connect the results of Tasks 2–9 to the audit committee's question.

Discuss:

- Which findings support the soundness of the compression, loss and optimiser choices
- Which findings raise concerns about the error profile
- How the risk of missed referable cases relates to the computed metrics

Your conclusion must be **traced to your computed results**.

Evidence Expected

A conclusion with each point linked to a specific computed result.

DAY-4 DESIGN CHALLENGE

TASK 11 – Analyse the Impact of Low Prevalence 8 Marks

Changed Requirement

RetinaCare will be deployed in a district where the prevalence of referable retinopathy is **2%** instead of 10%. A **missed referable case is ten times as costly** as an unnecessary referral. Sensitivity and specificity remain 90% and 95%.

Re-evaluate your original design in response to these changed operating conditions. You may **RETAIN → MODIFY → or REDESIGN** your original solution.

A. Impact Analysis – 2 Marks

Discuss:

- Which of your earlier results depend on prevalence and which do not
- Why sensitivity and specificity are unchanged while the posteriors change

Evidence Expected

A clear separation of prevalence-dependent and prevalence-independent quantities.

B. Re-examination – 2 Marks

Discuss:

- P(Referable | Positive) and P(Referable | Negative) at 2% prevalence
- The expected numbers of false referrals and missed cases per 1,000 patients, compared with 10% prevalence

Evidence Expected

Recomputed posteriors and expected counts with the change explained.

C. Analysis of Policy Options – 2 Marks

Examine the options for adjusting the screening policy:

- The minimum specificity needed for P(Referable | Positive) ≥ 0.5 at 2% prevalence
- How moving the decision threshold trades sensitivity against specificity
- The expected cost per 1,000 patients under the 10 : 1 cost ratio for each option

Evidence Expected

A derived specificity requirement and a cost comparison of the policy options.

D. Evidence-Based Conclusion – 2 Marks

State whether the policy should be retained, modified or redesigned, and trace it to:

- The recomputed probabilities and expected costs
- The cost of missed referable cases
- Practical constraints of rural screening

Your conclusion must follow logically from your calculations.

Evidence Expected

A conclusion whose every element is traced to the computed evidence.

MARK DISTRIBUTION

Task	**Marks**
Task 1 – Problem Decomposition	3
Task 2 – SVD-Based Compressibility Analysis	6
Task 3 – Hessian and Convexity Analysis	5
Task 4 – Information-Theoretic Comparison	5
Task 5 – Bayesian Decision Analysis	4
Task 6 – Gradient Descent Convergence Analysis	5
Task 7 – Momentum and Stability Analysis	4
Task 8 – Bias–Variance Decomposition	4
Task 9 – Error Profile Analysis	3
Task 10 – Integrated Analysis	3
Subtotal	**42**
Task 11 – Day-4 Design Challenge	**8**
TOTAL	**50**

Task 1 – Break Down the Audit Problem (3 marks)

The audit can be decomposed into six linked mathematical components:

Data item	**Concept and formula**	**Model stage**	**Assumption / dependency**
D1: $A = [[3,0],[4,5]]$	SVD: $A = U\Sigma V^T$; singular values from eigenvalues of A^TA	Representation / compression	Matrix values represent features; retained squared singular values measure energy.
D2: $J(w)=\frac{1}{2}w^THw-b^Tw$	Gradient $\nabla J=Hw-b$; Hessian $\nabla^2 J=H$; eigenvalues and condition number	Loss / optimisation	H is symmetric; its eigenvalues govern curvature and gradient-descent stability.
D3: p, q_1, q_2	Entropy $H(p)=-\Sigma p_i \log_2 p_i$; cross-entropy; KL divergence	Probabilistic output	p is the reference distribution; q distributions sum to 1.
D4: prevalence 10%, sensitivity 90%, specificity 95%	Bayes' rule and predictive values	Decision / screening	Prevalence is the prior; sensitivity and specificity are conditional probabilities.
D5: $y=10$, $\sigma^2=1$, predictions P and Q	Bias² + variance + noise variance	Prediction error / generalisation	Four predictions are treated as repeated estimates; population variance divides by 4.
D6: TP=40, FN=10,	Accuracy, error rate,	Evaluation	Referable is the

FP=5, TN=45	precision, recall, specificity, F1		positive class; confusion-matrix rows are actual classes.

Dependencies: D1 assesses compact representation; D2 assesses loss geometry and optimisation; D3 compares probability distributions; D4 converts screening characteristics into patient-level posteriors; D5 separates systematic error, variability and irreducible noise; D6 evaluates observed classification mistakes. Task 10 combines the findings and Task 11 changes prevalence and error costs.

Task 2 – Examine the Compressibility of the Feature Matrix (6 marks)

Given D1:

$A = [[3, 0], [4, 5]]$

Step 1: Calculate A^TA.

$A^TA = [[3,4],[0,5]]\ [[3,0],[4,5]] = [[25,20],[20,25]]$

Step 2: Solve the characteristic equation for eigenvalues λ.

$\det(A^TA - \lambda I) = (25-\lambda)^2 - 20^2 = 0$

$(25-\lambda)^2 = 400 \Rightarrow 25-\lambda = \pm 20$

$\lambda_1 = 45.0000;\ \lambda_2 = 5.0000$

Step 3: Find unit eigenvectors. For $\lambda_1=45$, an eigenvector is $[1,1]^T$; for $\lambda_2=5$, an eigenvector is $[1,-1]^T$. Normalising gives:

$v_1 = (1/\sqrt{2})[1,1]^T;\ v_2 = (1/\sqrt{2})[1,-1]^T$

Step 4: Singular values are square roots of eigenvalues of A^TA.

$\sigma_1 = \sqrt{45} = 6.7082;\ \sigma_2 = \sqrt{5} = 2.2361$

Step 5: Calculate U columns using $u_i = Av_i/\sigma_i$.

$Av_1 = (1/\sqrt{2})[3,9]^T \Rightarrow u_1 = (1/\sqrt{10})[1,3]^T$

$Av_2 = (1/\sqrt{2})[3,-1]^T \Rightarrow u_2 = (1/\sqrt{10})[3,-1]^T$

Therefore, one valid singular value decomposition is:

$U = (1/\sqrt{10})[[1,3],[3,-1]],\ \Sigma = [[3\sqrt{5},0],[0,\sqrt{5}]]$

$V = (1/\sqrt{2})[[1,1],[1,-1]],\ A = U\Sigma V^T$

Step 6: Rank-1 approximation $A_1 = \sigma_1 u_1 v_1^T$.

$A_1 = [[1.5000, 1.5000], [4.5000, 4.5000]]$

$A - A_1 = [[1.5000, -1.5000], [-0.5000, 0.5000]]$

Step 7: Frobenius reconstruction error.

$\|A-A_1\|_F = \sqrt{(1.5^2 + (-1.5)^2 + (-0.5)^2 + 0.5^2)}$

$\|A-A_1\|_F = \sqrt{5} = 2.2361$

Step 8: Percentage of energy retained.

Total energy = $\sigma_1^2 + \sigma_2^2 = 45 + 5 = 50$

Retained fraction = $\sigma_1^2/(\sigma_1^2+\sigma_2^2) = 45/50 = 0.9000$

Energy retained = 90.0000%; energy discarded = 10.0000%

Conclusion: $\sigma_1/\sigma_2=3.0000$, so the dominant singular value is three times the smaller one. Rank-1 compression retains 90% of squared energy with residual norm 2.2361. It is a plausible compact representation, but clinical validation must confirm that the discarded direction does not carry information important for detecting referable disease.

Task 3 – Analyse the Curvature of the Loss Surface (5 marks)

Given $H=[[4,1],[1,4]]$, $b=[5,5]^T$ and $J(w)=\frac{1}{2}w^THw-b^Tw$.

Step 1: Find eigenvalues.

$\det(H-\lambda I) = (4-\lambda)^2-1 = 0$

$4-\lambda = \pm 1 \Rightarrow \lambda_1=5.0000, \lambda_2=3.0000$

Step 2: Eigenvectors and curvature. $\lambda_1=5$ has eigenvector $[1,1]^T$; $\lambda_2=3$ has eigenvector $[1,-1]^T$. The [1,1] direction has higher curvature; [1,−1] has lower curvature.

Step 3: Positive definiteness. Both eigenvalues are positive, so H is positive definite. J(w) is strictly convex and has a unique global minimum.

Step 4: Condition number.

$\kappa_2(H)=\lambda max/\lambda min=5/3=1.6667$

A condition number of 1.6667 indicates moderate, not severe, curvature imbalance.

Step 5: Find and verify the minimiser.

$\nabla J(w)=Hw-b$; at a stationary point $Hw=b$

$H^{-1}=(1/15)[[4,-1],[-1,4]]$

$w^*=H^{-1}b=(1/15)[[4,-1],[-1,4]][5,5]^T=[1,1]^T$

$\nabla J(w^*)=[[4,1],[1,4]][1,1]^T-[5,5]^T=[0,0]^T$

Minimum loss: $J(w^*)=\frac{1}{2}([1,1]H[1,1]^T)-[5,5]\cdot[1,1]=\frac{1}{2}(10)-10=-5.0000$.

Task 4 – Differentiate the Models using Information Theory (5 marks)

Given $p=[0.5,0.25,0.125,0.125]$, $q_1=[0.25,0.25,0.25,0.25]$, $q_2=[0.4,0.3,0.2,0.1]$. Use base-2 logarithms.

Step 1: Entropy of p.

$H(p)=-\Sigma p_i \log_2 p_i$

$= -[0.5(-1)+0.25(-2)+0.125(-3)+0.125(-3)]$

$= 0.5000+0.5000+0.3750+0.3750 = 1.7500$ bits

Step 2: Entropy of uniform q_1.

$H(q_1)=-4(0.25 \log_2 0.25)=-4(0.25\times-2)=2.0000$ bits

Step 3: Cross-entropy $H(p,q_1)$.

$H(p,q_1)=-\Sigma p_i \log_2 q_{1i}=-\log_2(0.25)\Sigma p_i=2.0000$ bits

Step 4: KL divergence and decomposition.

$KL(p\|q_1)=\Sigma p_i \log_2(p_i/q_{1i})$

$= 0.5 \log_2 2 + 0.25 \log_2 1 + 0.125 \log_2 0.5 + 0.125 \log_2 0.5$

$= 0.5000 + 0.0000 - 0.1250 - 0.1250 = 0.2500$ bits

$H(p)+KL(p\|q_1)=1.7500+0.2500=2.0000=H(p,q_1)$

Step 5: Calculate class-wise terms and $KL(p\|q_2)$.

Class	**$p_i \log_2(p_i/q_{1i})$**	**$p_i \log_2(p_i/q_{2i})$**
No DR	0.5000	0.1610
Mild	0.0000	−0.0658
Moderate	−0.1250	−0.0848
Severe	−0.1250	0.0402
Total	0.2500	0.0507

$KL(p\|q_2)=0.1610-0.0658-0.0848+0.0402\approx0.0507$ bits

$H(p,q_2)=H(p)+KL(p\|q_2)=1.7500+0.0507=1.8007$ bits

q_2 is closer to p because $0.0507 < 0.2500$. No DR contributes the largest positive divergence term for q_2; Severe contributes a smaller positive term. Mild and Moderate terms are negative because q_2 assigns them more probability than p. Individual terms can be negative, but total KL divergence cannot be negative. A divergence of 0 means the distributions match exactly (for all classes with positive probability).

Task 5 – Analyse the Screening Decision using Bayes' Rule (4 marks)

D4: $P(R)=0.10$, $P(\text{not } R)=0.90$, sensitivity $P(+|R)=0.90$, and specificity $P(-|\text{not } R)=0.95$. Thus $P(-|R)=0.10$ and $P(+|\text{not } R)=0.05$.

Branch	**Calculation**	**Joint probability**
True positive	$P(R)P(+\|R)=0.10\times0.90$	0.0900
False negative	$P(R)P(-\|R)=0.10\times0.10$	0.0100

False positive	P(not R)P(+|not R)=0.90×0.05	0.0450
True negative	P(not R)P(−|not R)=0.90×0.95	0.8550

Positive posterior / PPV:

P(R|+)=P(+|R)P(R)/P(+)

=0.0900/(0.0900+0.0450)=0.0900/0.1350=0.6667

Probability of referable disease after a negative result:

P(R|−)=0.0100/(0.0100+0.8550)=0.0100/0.8650=0.0116

Sensitivity is P(+|R), while PPV is P(R|+); they condition on different events. PPV is lower because false positives arise from the much larger non-referable population. A positive screen alone should not determine treatment; it should prompt clinical assessment or confirmatory testing.

Task 6 – Trace the Convergence of Gradient Descent (5 marks)

Gradient: $\nabla J(w)=Hw-b$. Update: $w_{t+1}=w_t-\eta\nabla J(w_t)$, $\eta=0.1$, $w_0=[0,0]^T$. Symmetry means $w=[x,x]^T$ and $J=5x^2-10x$.

Iteration 1:

$\nabla J(w_0)=[-5,-5]^T$; $w_1=[0,0]-0.1[-5,-5]=[0.5000,0.5000]^T$

$J(w_1)=5(0.5^2)-10(0.5)=-3.7500$

Iteration 2:

$\nabla J(w_1)=[4(0.5)+0.5-5,\ 0.5+4(0.5)-5]=[-2.5,-2.5]$

$w_2=[0.5,0.5]-0.1[-2.5,-2.5]=[0.7500,0.7500]^T$

$J(w_2)=5(0.75^2)-10(0.75)=-4.6875$

Iteration 3:

$\nabla J(w_2)=[-1.25,-1.25]$; $w_3=[0.7500,0.7500]-0.1[-1.25,-1.25]$

$w_3=[0.8750,0.8750]^T$; $J(w_3)=5(0.875^2)-10(0.875)=-4.9219$

Step	**w**	**J(w)**	**Distance to $w^*=[1,1]^T$**
0	[0.0000,0.0000]	0.0000	1.4142
1	[0.5000,0.5000]	−3.7500	0.7071
2	[0.7500,0.7500]	−4.6875	0.3536
3	[0.8750,0.8750]	−4.9219	0.1768

The distance halves at each step because the error lies in the $\lambda=5$ eigenvector direction and its multiplier is $1-\eta\lambda=1-0.1\times5=0.5000$. The loss approaches the minimum -5.0000.

Task 7 – Differentiate Momentum from Plain Gradient Descent (4 marks)

Use $v_{t+1}=\beta v_t-\eta\nabla J(w_t)$, $w_{t+1}=w_t+v_{t+1}$, with $\beta=0.9$, $\eta=0.1$, $v_0=0$ and $w_0=0$.

Iteration 1:

$\nabla J(w_0)=[-5,-5]$; $v_1=0.9[0,0]-0.1[-5,-5]=[0.5000,0.5000]$

$w_1=[0.5000,0.5000]$; $J(w_1)=-3.7500$

Iteration 2:

$\nabla J(w_1)=[-2.5000,-2.5000]$

$v_2=0.9[0.5,0.5]-0.1[-2.5,-2.5]=[0.7000,0.7000]$

$w_2=[0.5,0.5]+[0.7,0.7]=[1.2000,1.2000]$

$J(w_2)=5(1.2^2)-10(1.2)=-4.8000$

Momentum overshoots $w^*=[1,1]^T$ because accumulated velocity persists while the gradient magnitude is shrinking.

For plain gradient descent on a positive-definite quadratic, stability requires $0<\eta<2/\lambda_{max}$. Here $\lambda_{max}=5$:

$0 < \eta < 2/5 = 0.4000$

At $\eta=0.5$, plain gradient descent fails: $w_1=0-0.5[-5,-5]=[2.5000,2.5000]$. Then $\nabla J(w_1)=[7.5000,7.5000]$ and $w_2=[2.5,2.5]-0.5[7.5,7.5]=[-1.2500,-1.2500]$. The error multiplier in the $\lambda=5$ direction is $1-\eta\lambda=-1.5000$, so the iterates alternate across the minimum and diverge. This bound is for plain gradient descent; momentum has a different joint stability condition involving η, β and the eigenvalues.

Task 8 – Decompose the Prediction Error (4 marks)

Given target $y=10$ and $\sigma^2=1$. For each four-value set, use population variance (divide by $n=4$).

Model P: predictions [8,9,8,9].

Mean = (8+9+8+9)/4 = 8.5000; bias = 8.5000−10 = −1.5000

Bias² = (−1.5000)² = 2.2500

Variance = [(-0.5)²+(0.5)²+(-0.5)²+(0.5)²]/4 = 0.2500

Expected squared error = Bias²+Variance+σ² = 2.2500+0.2500+1.0000 = 3.5000

Model Q: predictions [12,7,11,8].

Mean = (12+7+11+8)/4 = 9.5000; bias = 9.5000−10 = −0.5000

Bias² = (−0.5000)² = 0.2500

Variance = [(2.5)²+(−2.5)²+(1.5)²+(−1.5)²]/4 = 4.2500

Expected squared error = 0.2500+4.2500+1.0000 = 5.5000

Model	Bias²	Variance	Noise σ²	Expected squared error
P	2.2500	0.2500	1.0000	3.5000
Q	0.2500	4.2500	1.0000	5.5000

P is dominated by bias but has low variance; Q has lower bias but much higher variance. P is preferable under the requested decomposition because its total expected squared error is 3.5000 versus 5.5000. The observed mean squared prediction errors before adding noise are 2.5000 for P and 4.5000 for Q.

Task 9 – Examine the Error Profile of the Classifier (3 marks)

From D6: TP=40, FN=10, FP=5, TN=45, N=100.

Metric	Formula	Result
Accuracy	(TP+TN)/N = 85/100	0.8500 = 85.0000%
Error rate	(FP+FN)/N = 15/100	0.1500 = 15.0000%
Precision	TP/(TP+FP)=40/45	0.8889 = 88.8889%
Recall / sensitivity	TP/(TP+FN)=40/50	0.8000 = 80.0000%
Specificity	TN/(TN+FP)=45/50	0.9000 = 90.0000%
F1-score	2TP/(2TP+FP+FN)=80/95	0.8421 = 84.2105%

False negatives (10) are twice as frequent as false positives (5). False negatives can delay assessment for referable retinopathy; false positives can cause anxiety, travel and unnecessary referrals. Accuracy alone conceals error type and clinical cost, so recall, specificity, precision and the false-negative cost must also be considered.

Task 10 – Relate the Findings to the Audit Question (3 marks)

- Compression: Task 2 retained 90.0000% of matrix energy, with residual Frobenius norm 2.2361. This supports compactness, but not clinical safety by itself.
- Loss and optimisation: Task 3 gave positive eigenvalues 5.0000 and 3.0000, condition number 1.6667 and unique minimiser $[1,1]^T$. Gradient descent converges at $\eta=0.1$; $\eta=0.5$ is unstable for plain gradient descent, and momentum can overshoot.
- Distribution comparison: Task 4 found $KL(p\|q_2)=0.0507$ bits, below $KL(p\|q_1)=0.2500$ bits, so q_2 is closer to p.
- Bayesian decision: At 10% prevalence, D4 gives PPV=66.6667% and $P(R|-)=1.1561\%$.

- Observed classifier profile: D6 gives recall 80.0000%, so 20.0000% of actual referable cases in that matrix are missed; there are 10 false negatives versus 5 false positives.

The loss geometry and optimiser are mathematically sound under the stated step size, and compression is efficient by energy. Nevertheless, clinical approval must focus on false negatives, local prevalence, calibration, threshold selection and external validation, rather than accuracy or compression energy alone.

Task 11 – Day-4 Design Challenge: Low Prevalence and Unequal Costs (8 marks)

A. Impact Analysis (2 marks)

The prevalence changes from 10% to 2%; sensitivity remains 90% and specificity remains 95%. Sensitivity P(+|R) and specificity P(−|not R) are conditional rates stipulated to stay constant. SVD, entropy and Hessian results do not depend on prevalence. Predictive values, expected counts and costs do depend on prevalence.

Bayes' rule changes the posterior because the prior probability of disease is smaller. A larger non-referable population can generate many false positives even when specificity is unchanged.

B. Re-examination at 2% prevalence (2 marks)

For 1,000 patients: 2% are referable → 20 referable and 980 non-referable.

TP = 20×0.90 = 18; FN = 20×0.10 = 2

TN = 980×0.95 = 931; FP = 980×0.05 = 49

Expected count per 1,000	**At 10% prevalence**	**At 2% prevalence**
Referable patients	100	20
Non-referable patients	900	980
True positives	90	18
False negatives	10	2
True negatives	855	931
False positives	45	49

Positive posterior at 2%:

P(R|+) = TP/(TP+FP) = 18/(18+49) = 18/67 = 0.2687 = 26.8657%

Disease probability after a negative result:

P(R|−) = FN/(FN+TN) = 2/(2+931) = 2/933 = 0.0021 = 0.2144%

At 10% prevalence, PPV=90/(90+45)=0.6667 and P(R|−)=10/(10+855)=0.0116. At 2%, PPV falls to 26.8657%, while P(R|−) falls to 0.2144%. False referrals increase from 45 to 49 per

1,000 because there are many more non-referable patients; missed cases fall from 10 to 2 per 1,000 because prevalence is lower, not because sensitivity improved.

C. Policy Options and Expected Costs (2 marks)

1. Minimum specificity needed for PPV ≥ 0.5. PPV is at least 0.5 when true positives are at least as numerous as false positives.

Sensitivity×prevalence ≥ (1−specificity)×(1−prevalence)

0.90×0.02 ≥ (1−Sp)×0.98

0.0180 ≥ 0.98(1−Sp)

1−Sp ≤ 0.0180/0.98 = 0.0184

Sp ≥ 1−0.0183673 = 0.9816327 = 98.1633%

Thus, if sensitivity remains 90%, specificity must be at least 98.1633% to achieve PPV ≥ 50%. This is a mathematical requirement, not evidence that a real threshold can reach it without changing sensitivity.

2. Cost model: let one false positive cost 1 unit and one false negative cost 10 units.

Expected cost per 1,000 = 10×FN + FP

At prevalence 2%: Cost = 10×[20(1−sensitivity)] + 980×(1−specificity)

Scenario	**Sensitivity**	**Specificity**	**FN / FP per 1,000**	**Cost units per 1,000**
Current stated operating point	90.0000%	95.0000%	2 / 49	10×2+49 = 69
Illustration: raise specificity to 98.1633%, assume sensitivity unchanged	90.0000% (assumed)	98.1633%	2 / about 18	about 38
Redesigned threshold / two-stage pathway	Must be measured	Must be measured	Depends on measured rates	10×FN+FP

The illustrative second row is not a guaranteed achievable threshold: exact cost depends on the actual paired sensitivity and specificity. The assessment gives no ROC curve or

alternative threshold results, so precise costs for a real redesigned policy cannot be calculated from the supplied data. If sensitivity drops, false negatives and their tenfold cost rise.

D. Evidence-Based Conclusion (2 marks)

Recommendation: MODIFY the policy and validate candidate thresholds locally before deployment. Do not automatically retain the current threshold as the only rule, and do not claim a redesigned threshold is proven without measured operating-point data.

- At 2% prevalence, PPV is 26.8657%; about 73.1343% of positive screens would be false positives under the stated rates.
- The current expected cost is 69 units per 1,000 patients using the 10:1 false-negative-to-false-positive cost ratio.
- Specificity of at least 98.1633% is required for PPV $\geq$ 50% if sensitivity remains 90%. Under that assumed operating point, cost would be about 38 units per 1,000; feasibility is not established by the question data.
- Because a missed case costs ten times more, increasing specificity must not be allowed to cause an unmeasured or unacceptable sensitivity loss. Choose a threshold using local validation, a clinically agreed sensitivity floor and the explicit cost function.
- In rural screening, positive screens should lead to a feasible confirmatory/referral pathway. A negative result should not override clinical concern or poor image quality. Monitor missed cases, referral completion, calibration and subgroup performance.

The supplied evidence supports modifying the policy and validating it locally. Exact performance and costs for a redesigned threshold cannot be stated without sensitivity and specificity measured together at candidate thresholds.

Overall Conclusion

The audit shows that rank-1 compression retains 90.0000% of matrix energy; the quadratic loss is strictly convex with a unique minimiser; gradient descent converges with $\eta=0.1$; and q_2 is closer to the reference distribution than q_1. However, a low prevalence of 2% reduces PPV to 26.8657% at the given test characteristics, and the cost ratio makes false negatives particularly important. The policy should be modified and validated using local prevalence, measured threshold trade-offs and explicit penalties for missed referable disease. Mathematical convergence and compression are necessary engineering properties, but do not by themselves establish clinical s